Baldivis Secondary College

Year 9

Mini Test 1

Number and Algebra

Total Marks: 32 marks

Name: Answers

Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Questions 1-8 are multiple choice one mark each.
Circle the correct answer.

1. 4^3 in expanded form is:

A 64 **B** $4 \times 4 \times 4$ **C** $3 \times 3 \times 3 \times 3$

D 4^3 **E** 12

2. $2^{12} \div 2^7$ is equal to:

A 4 **B** 8 **C** 16

D 25 **E** 32

3. $2x^7 \times 7x^3$ is equivalent to:

A $14x^{21}$ **B** $14x^{10}$ **C** $14x^4$

D $9x^{10}$ **E** $9x^{21}$

4. $-2x^0 + (9x)^0$ simplifies to:

A -2 **B** -1 **C** 0

D 2 **E** 7

5. $\frac{6a^7b^5}{3a^2b^8}$ simplifies to:

A $\frac{2a^5}{b^3}$ **B** $\frac{b^5}{2a^3}$ **C** $\frac{2a^3}{b^5}$

D $\frac{2b^3}{a^5}$ **E** $\frac{a^5}{2b^3}$

6. When we expand brackets, we use the distributive law. Which of the following does *not* represent the distributive law?

A $a(b + c) = ab + ac$ **B** $-a(b + c) = -ab - ac$ **C** $-a(b - c) = -ab - ac$

D $-a(b - c) = -ab + ac$ **E** $a(b - c) = ab - ac$

7. Which of the following expressions is equivalent to $2(5m + 1)$?

A $7m + 1$ **B** $10m + 1$ **C** $10m + 2$

D $12m$ **E** $11m$

8. Which of the following expressions is equivalent to $5c + 10$?

A $2c \times 3c + 10$ **B** $5c + c + 10$ **C** $15c$

D $c + 16 + 4c - 6$ **E** $2c + 16 - 3c - 6$

9. Evaluate

a. 2^4 ~~2^3 3^1 2^4 16~~ $3^3 = 27$

b. 7^0 $= 1$

c. 5^{-2} $\frac{1}{25}$

d. $(3y^0)^3$ ~~27~~ 8

[4 marks]

10. Simplify the following expressions

a. $6 \times m \times m \times m$

$6m^3$

b. $3bc \times 2ab$

~~$6b^2$~~ $6ab^2c$

c. $2p - 3p^2 + 7p + 5p^2 - 3$ $9p + 2p^2 - 3$

d. $t^3 \times t^5$

t^8

e. $(n^3)^4$

n^{12}

f. $3a^4b^5 \times 7a^7b$

$21a^{11}b^6$

g. $\dfrac{3(e^4)^5 \times e^2}{9e^{-3}}$

$\frac{e^{25}}{3}$ (2marks)

[8 marks]

11. Which two algebraic expressions are equivalent?(Circle your choices)

A $7x + 7$ **B** $7x + 7y$ **C** $7xy$ **D** $x + 7y$ **E** $7(x + y)$

[1 mark]

12. Expand the following and simplify when possible.

a. $5(x - 1)$ $5x - 5$.

b. $2x(5 - x)$ $10x - 2x^2$.

c. $10x - 3(2x + 1)$ $10x - 6x - 3$.
$4x - 3$.

d. $5(x - 1) + 2(3x - 7)$ $5x - 5 + 6x - 14$
$= 11x - 19$.

[6 marks]

13. Fully factorise the following expressions

a. $6y + 21$ $3(2y + 7)$.

b. $5x^2 - 15x$ $5x(x - 3)$.

c. $-4p^2 - 2p$ $-2(2p + 1)$.

d. $3a^2b - 9a^3b^2$ $3a^2b(1 - 3ab)$.

[5 marks]

End of test